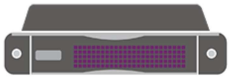
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Hardware Security Modules

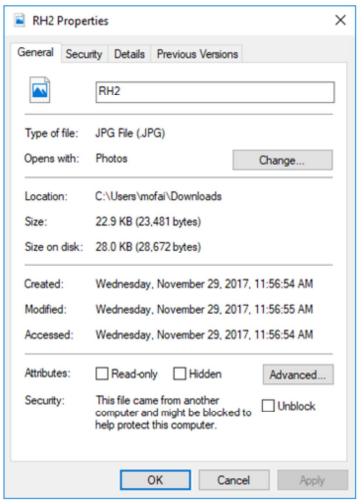




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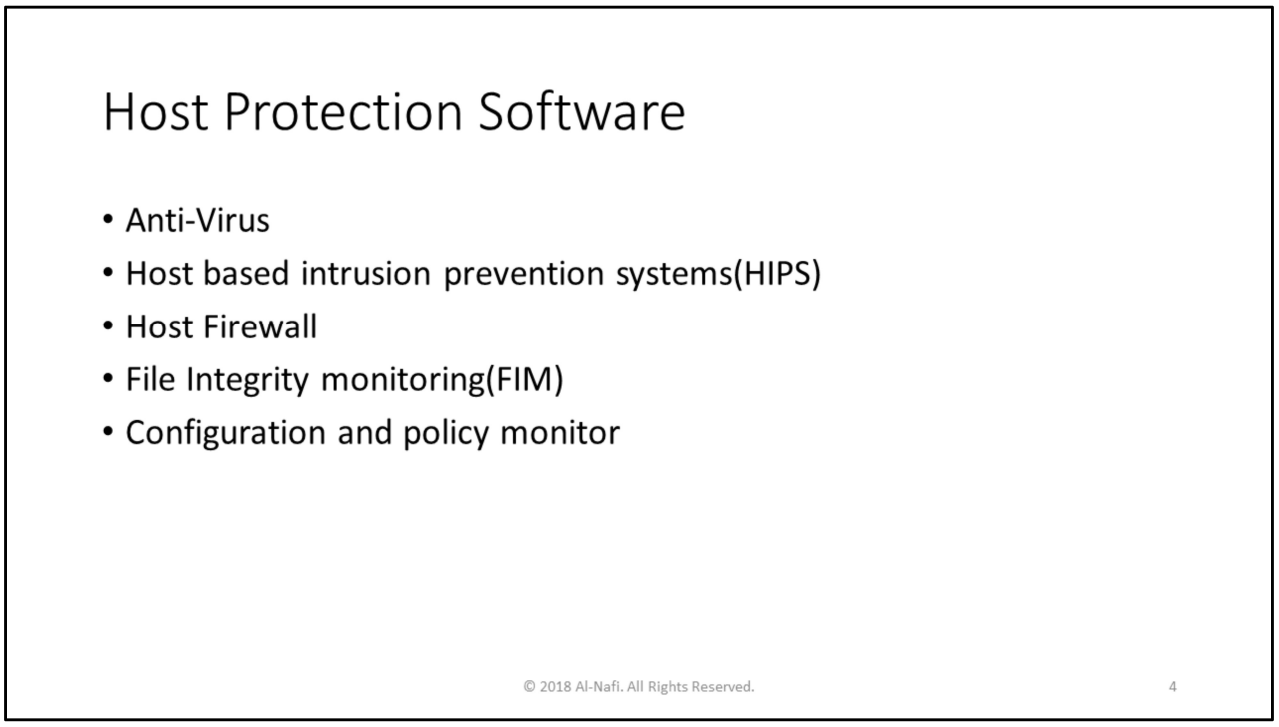
Hardware components may be used to provide security services to the system. A common example is the Trusted Platform Module (TPM) that is provided by or available as an option on most major device manufacturers. The TPM is a hardware module that includes a secure storage container and a cryptographic processor with some cryptographic functions. It is typically used to securely generate and store cryptographic keys or provide secure storage of small data sets.
The most common use for a TPM is to generate and store cryptographic keys associated with file system or drive encryption mechanisms. Since the keys are stored within the dedicated hardware module, they are extremely difficult to extract when the system is powered down. They are only exposed at certain points during the boot process that are difficult to monitor prior to the OS being functional and taking over the role of protecting the keys. Other hardware security modules exist for specialty functions and may be added to systems or used as peripheral devices for special security functions.

File System attributes



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Modern files systems store security attributes, or permissions, associated with files as an integral part of the file system. This enables advanced security models to be employed in practical systems and ensures easy association of security attributes with individual files. Some file systems include journaling that protects file integrity by ensuring that incomplete disk operations are identified and completed.



The following are examples of host protection software that may be installed at the system level to provide additional protections beyond those built into the OS and system architecture. Some may be available as OS components but must typically be enabled and configured for full function. In other cases, third-party software suites may be used to provide these functions.

Antivirus: Protects against viruses and malicious code by checking files against a list of known malware. Many products also include a heuristics function that allows them to identify malware that is not in their database based on software behavior.

Host-based intrusion prevention system (HIPS): HIPS provides monitoring of system communications and performs a similar function to a network-based intrusion prevention system (NIPS) within a specific host.

Host firewall: Blocks inbound or outbound communications from the host based on a defined rule set. Some host firewalls allow applications to dynamically configure the firewall to allow on-demand communications when necessary.

File integrity monitoring (FIM): Creates a known baseline of all files on a system, typically using a cryptographic hashing mechanism to create unique signatures for each file. It can then compare files against the known baseline periodically or when the files are loaded into memory for use.

Configuration and policy monitor: A configuration or policy monitor provides oversight to ensure defined system configurations or policies are correctly configured and not improperly modified. It may also report system status or compliance to an enterprise tool.

Vulnerabilities of Security Architectures, Designs, and Solutions Elements

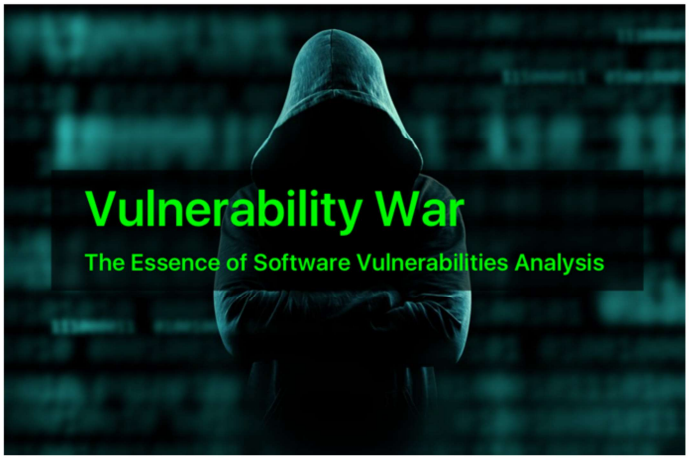
- Systems hardware
- Systems code
- System misuse opportunities
- System communications

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This topic introduces some common vulnerabilities and mitigation approaches that are common among most system types. It then presents typical vulnerabilities and mitigation approaches for various system types. The vulnerabilities and mitigations are not intended to be comprehensive for each system type and represent the most common issues and solutions associated with the system type. For each system type, consider which common vulnerabilities might exist in the various system components in addition to the system specific vulnerabilities. In particular consider how common vulnerabilities might exist in the
following:

- System hardware
- System code
- System misuse opportunities
- System communications

Systems hardware



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Systems code



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System misuse opportunities



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System communications



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Top Threats and their Mitigations

- Hacking
- Social Engineering
- Malware distribution
- Phishing

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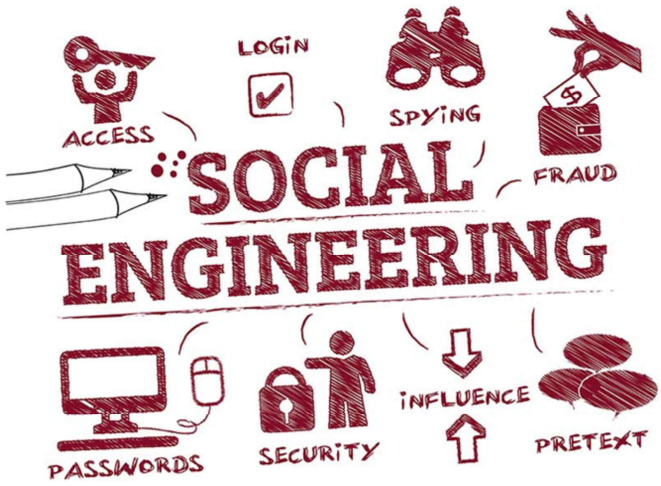
Hacking



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Hacking: Human action attempting various permutations of actions to defeat or bypass system protections or system security.

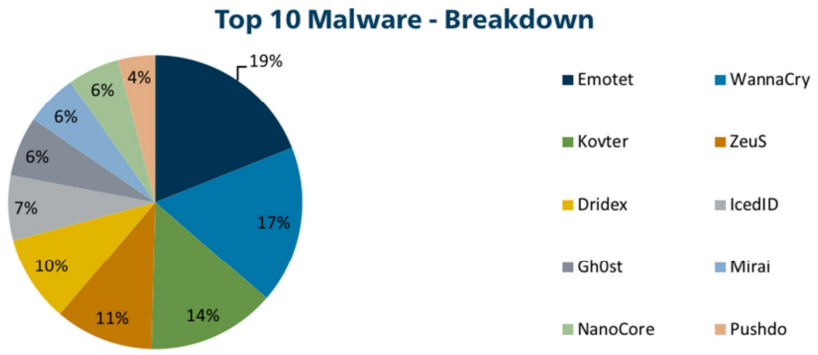
Social engineering



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Social engineering: Attempting to gain information or access by impacting human behavior or process. Generally implemented through human interaction but may be message or communication based.

Malware distribution



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Malware distribution: Manual or automated distribution of malware. May be targeted, untargeted, or the result of self replicating malware moving autonomously.

Phishing



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Phishing: Attempting to gain information or access by sending messages (e.g., email) that seem to be legitimate but are not. May be combined with types of social engineering or malware distribution.

Top Mitigations

- Know what you have
- Patch and manage me what you have
- Assess/monitor/log
- Educate users

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